Learning public health informatics with interactive notebooks

Field demonstration



Laila's district surveillance report

Which district needs follow-up first?

District	Previous week	Current week
Kambia	2	7
Port Loko	3	4
Bombali	4	3

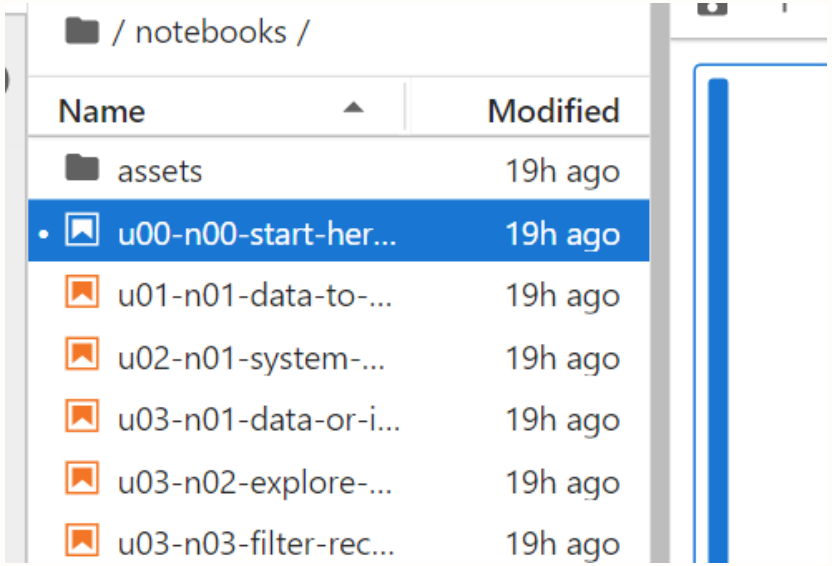
What would you verify before acting on these reports?

Invented reported case counts from the course

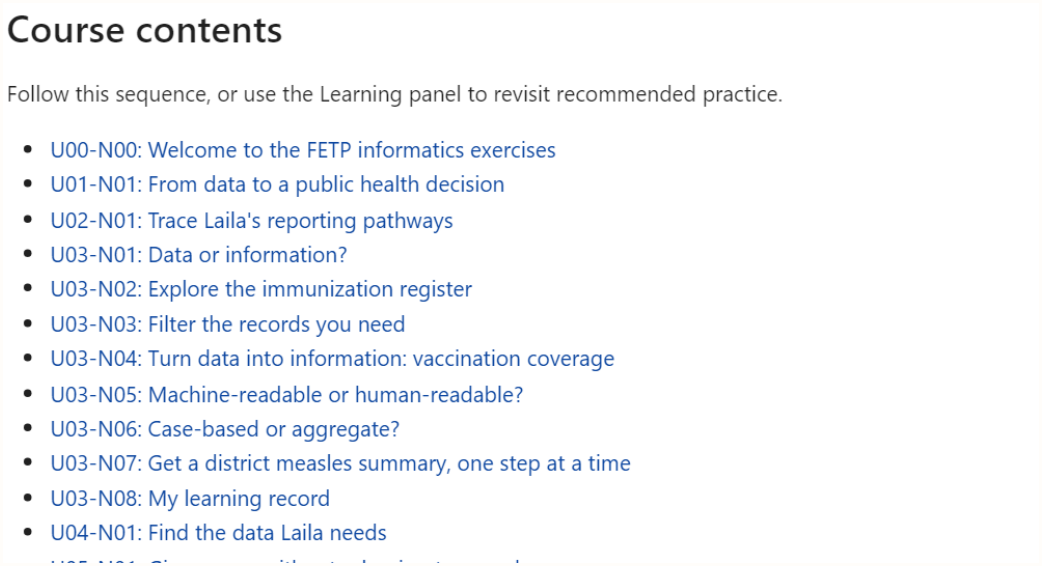
JupyterLite: where to start

A computing workspace that runs in your browser

1 Open Notebook 00



2 Choose an activity link



Read the activity instructions before running cells.

Open the course start page

A computational narrative

The question

Which reported change needs follow-up?

The calculation

```
previous_week = 2
current_week = 7
current_week - previous_week
```

The result

5 additional reported cases

Interpretation: verify the reports and investigate the signal.

The tools supporting the activity

Tool	What the learner uses it for
Jupyter notebook	Read the case, run steps, inspect results
JupyterLab interface	Open files and use notebook controls
JupyterLite	Work through the course in a browser
Python and its libraries	Calculate summaries and draw charts
Course extension	Get feedback and suggested activities

Learning through a notebook

01	**Predict**	What do you expect to find?
02	**Try**	Run a step or choose an answer.
03	**Inspect**	Compare the result with the data.
04	**Explain**	Describe the reasoning in your own words.
05	**Revise**	Use feedback to improve the answer.
06	**Apply**	Work through a different example.

Adaptive learning support

Submit an answer and inspect the feedback.

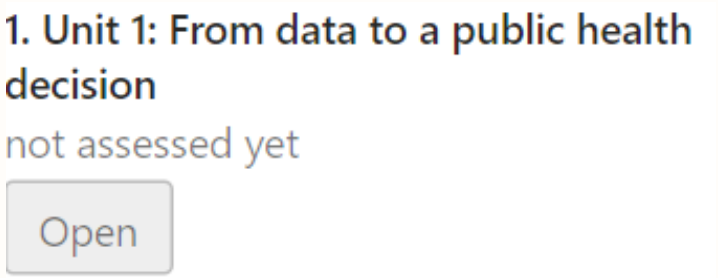
Table A: Incorrect. Each row describes one case. A number or age-group label can still describe an individual case.

Use authored hints to work through a difficulty.

The Learning panel uses recorded results and help to suggest what to try next.

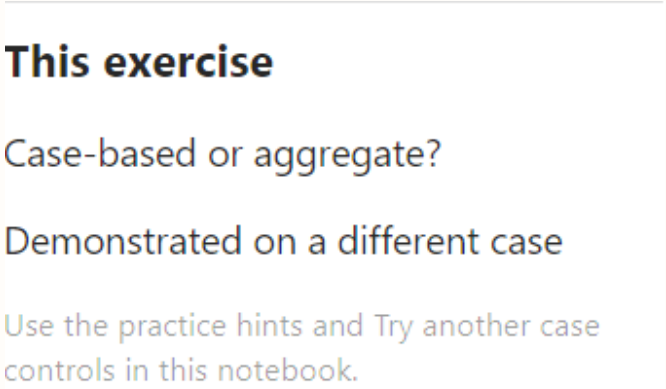
The Learning panel

Suggested activities



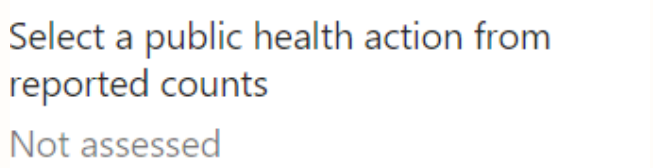
Open a recommended activity.

Current exercise



See support for the current task.

Recorded progress



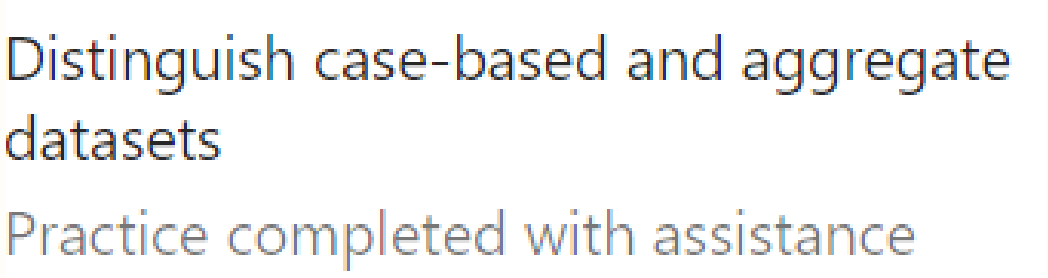
Review progress on course tasks.

Suggestions reflect your recorded attempts and help used.

Practice and a different case

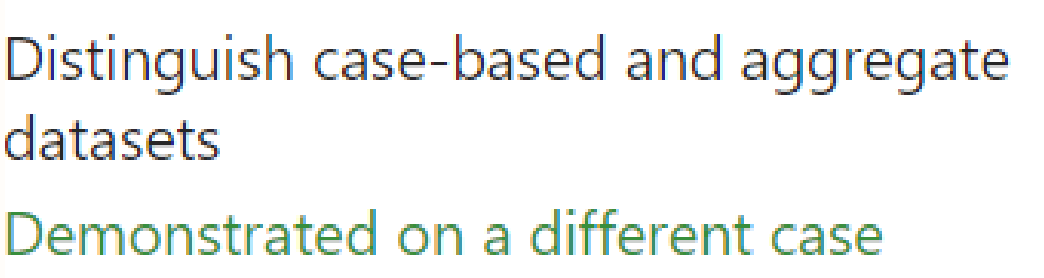
Practice

Use feedback and hints.
Revise your classifications.



A different case

Apply the same reasoning
to unfamiliar tables.



These labels describe performance on this classification task.

Walkthrough: choices and feedback

Run a selected code cell with Shift+Enter



Table A: Individual case register

Invented example rows

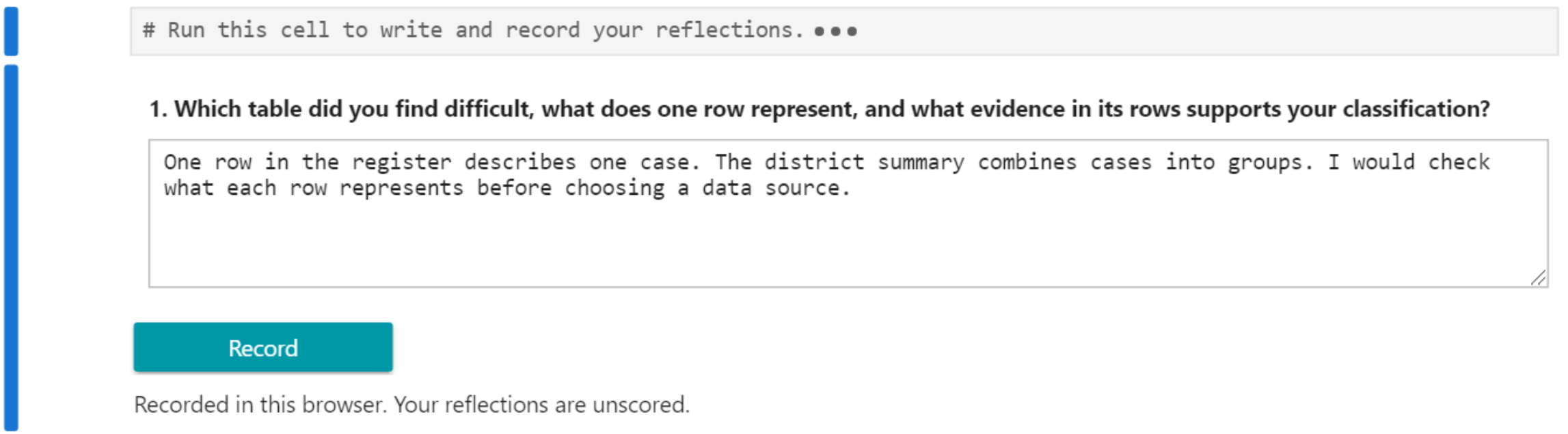
Case_ID	Age	District	Diagnosis
P01	8	Kambia	Suspected measles
P02	12	Kambia	Suspected measles

- ○ Case-based
- ○ Aggregate

1. Run the setup cell
2. Choose an answer for every table
3. Submit and read feedback
4. Try another case

Open the live activity

Your learning record and reflections



Select Record to save your reflection.

Review attempts and progress in My learning record.

Work stays in this browser for this course site.

Learning with a facilitator

01 What does one row in this table represent?

02 What changed in your answer after the feedback?

03 What would you verify before using this report?

Epidemiologists as course authors

Use your field experience to shape the next version.

An example from this course

Give a reflection question
a working response field
and a Record button.

An example request to an AI assistant

"Inspect this activity. Add a response field and a Record button. Show how to check that the saved response returns after reload."

Work with invented data and an approved AI tool.

A typical improvement workflow

01	**Identify**	Name the gap and the learner goal.
02	**Plan**	Review a small change proposed with AI.
03	**Build**	Revise one activity using invented data.
04	**Try**	Check correct, incorrect, and empty answers.
05	**Review**	Check meaning, usability, and saved work.
06	**Publish**	Share the reviewed version and verify it live.

Reviewing an AI-assisted contribution

Public health meaning	Do the categories answer the intended question?
Results and feedback	Do the calculation, label, and explanation agree?
Supporting sources	Does each reference support the claim?
A colleague's experience	Can someone complete the activity without coaching?

Contribute the revised activity with a short change and review note.

Questions for the field test

Could you find where to start?

Did the feedback help you explain your answer?

What improvement would you contribute?

Choose one activity to review and improve

adaptive-learning-network.github.io/jupyterlite-prototype